Doga Tekin

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AI researcher & engineer with 5+ years of experience, working on LLM-based agents and relational deep learning. Academically excellent + a track record of building impactful AI systems for support, cybersecurity, finance, and more.

Education

EPFL 2018 – 2021
Computer Science MSc | GPA: 5.85 / 6.00 *Lausanne, CH*
- Graduated **1st among 112 students** in the CS department.

Bilkent University 2014 – 2018
Electrical & Electronics Engineering BSc | GPA: 3.79 / 4.00 *Ankara, TR*
- Received a scholarship for ranking in the **top 0.1% in Turkey** on the entrance exam.

Work Experience

Oracle Labs & Oracle Database AI Research 2021 – Present
Principal Member of Technical Staff (*2026–Present*) *Zurich, CH*
- Leading the development of an LLM-based agent for customer support request analysis and escalation.
  - It can correctly identify 5x more critical cases than the existing manual process.
  - Will process 1M+ support requests annually, saving time and effort for >3000 support associates.
- Training Relational Graph Transformers to generate embeddings for structured-entity similarity search.
  - Outperforms tabular / relational foundation model baselines such as TabICL and KumoRFM.

Senior Member of Technical Staff (*2023–2026*)
- Led the development of an LLM-based agent that analyzes the cybersecurity of Oracle Cloud tenancies.
  - Found several concrete vulnerabilities in production tenancies that were fixed thanks to the agent.
- Developed a transformer-based method that transforms tables into graphs for use in Oracle Finance products.
  - Filed a pending patent application for this method.
- Mentored and supervised 20+ interns on NLP, Graph ML, LLM, RL, and cybersecurity topics.

Member of Technical Staff (*2021–2023*)
- Improved Graph ML explainability features in Oracle's graph toolkit, PGX.
- Used feature engineering and AutoML to train accurate trajectory prediction models for a major Oracle customer.
- Analyzed and visualized large-scale data to aid leadership decision-making using PySpark and SQL.

Accenture Labs 2020 – 2020
Research Intern *Dublin, IE*
- Implemented evolutionary hyperparameter optimization algorithms in Accenture's graph library AmpliGraph.
  - Tested on public knowledge graph benchmarks to prove this can help users train better embedding models.

Publications

NeurIPS 2019 OptRL Workshop
Reinforcement Learning with Langevin Dynamics – K. Parameswaran, **D. Tekin**, P. Rolland, V. Cevher
- Implemented a novel exploration method for reinforcement learning algorithms using TensorFlow.
- Conducted extensive testing on public RL benchmarks to produce results for the paper.

Skills

- **ML & AI**: LLMs, Agents, NLP, Graph ML, Relational Learning, Reinforcement Learning
- **Frameworks & Libraries**: PyTorch, Hugging Face, scikit-learn, NumPy, pandas, PySpark
- **Programming**: Proficient in Python, SQL, JavaScript. Used some Java and C++.

Personal Projects

Blog, Tools, Games https://www.dogatekin.com
- Blog about topics ranging from drawing with code to using ML frameworks like JAX for language modeling.
- Building and sharing small tools & games.